

Donghwi Jung

Autonomous Robot Intelligence Lab.
Smart City Engineering Major, Department of Civil & Environmental Engineering
Seoul National University
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Research Interests

Robotics, SLAM, Spatial Understanding, Vision-Language Navigation, Human-Robot Collaboration

Education

Seoul National University Sep. 2020 - Expected Feb. 2026
Ph.D. in Civil & Environmental Engineering
Advisor: Prof. Seong-Woo Kim

Seoul National University Mar. 2008 - Feb. 2017
B.S. in Food Science & Biotechnology / B.A. in Business Administration

Publications

Donghwi Jung, Jihun Moon, Jiyang Lee, DongYeop Shin, Jungjae Lee, Jinhee Kim, Seong-Woo Kim, “*Beyond Destinations: Instruction-Aware Graph Path Planning and Navigation with OpenStreetMap*,” IEEE International Conference on Robotics and Automation (ICRA), 2026. (under review)

Donghwi Jung, Keonwoo Kim, Seong-Woo Kim, “*GOTPR: General Outdoor Text-based Place Recognition Using Scene Graph Retrieval with OpenStreetMap*,” IEEE Robotics and Automation Letters, May 2025.

Chan Kim*, Keonwoo Kim*, Mintaek Oh, Hanbi Baek, Jiyang Lee, **Donghwi Jung**, Soojin Woo, Younkyung Woo, John Tucker, Roya Firoozi, Seung-Woo Seo, Mac Schwager, and Seong-Woo Kim, “*E2Map: Experience-and-Emotion Map for Self-Reflective Robot Navigation with Language Models*,” IEEE International Conference on Robotics and Automation (ICRA), Atlanta, USA, May 2025.

Donghwi Jung, Andres Pulido, Jane Shin, Seong-Woo Kim, “*Point Cloud Structural Similarity-Based Underwater Sonar Loop Detection*,” IEEE Robotics and Automation Letters, March 2025.

Yoonkyu Yoo*, **Donghwi Jung***, Seong-Woo Kim, “*3D Operation of Autonomous Excavator based on Reinforcement Learning through Independent Reward for Individual Joints*,” arXiv preprint arXiv:2406.19848. (* indicates equal contribution)

Donghwi Jung, Jae-Kyung Cho, Younghwa Jung, Soohyun Shin, Seong-Woo Kim, “*LoRCOn-LO: Long-term Recurrent Convolutional Network-based LiDAR Odometry*,” International Conference on Electronics, Information, and Communication (ICEIC), 2023.

Donghwi Jung, Younghwa Jung, Woo-Sol Hur, Jae-Kyung Cho, and Seong-Woo Kim, “*Multi-Floor Mapping including the Elevator*,” Summer Annual Conference of the Institute of Electronics and Information Engineers, June 2022.

Byeong Yeon Ryu, Won Nyoung Park, **Donghwi Jung** and Seong-Woo Kim, “*Landmark Localization for Drone Aerial Mapping Using GPS and Sparse Point Cloud for Photogrammetry Pipeline Automation*,” International Conference on Electronics, Information, and Communication (ICEIC), Feb. 2022.

Projects

<i>Visual-Language Outdoor Navigation Using OpenStreetMap</i> Internal Laboratory Project Role - Navigation Algorithm Implementation, Data Acquisition and Experimentation, Team Leadership	Apr. 2025 - Sep. 2025
<i>Visual-Language Indoor Navigation Using Foundation Models</i> Internal Laboratory Project Role - 2D/3D Mapping, Virtual Environment Implementation, Visual-Language Map Generation, and Localization	Apr. 2024 - Sep. 2024
<i>Development of strong autonomous intelligence for unmanned vehicles</i> R&D Project funds from the Agency for Defense Development, Republic of Korea Role - LiDAR Odometry Estimation	Jan. 2022 - July 2023
<i>Location estimation method in pretreatment and painting factory using inertial sensors</i> Industrial Cooperative Project funds from Daewoo Shipbuilding & Marine Engineering Co., Ltd Role - Inertial Odometry Estimation	Sep. 2022 - Feb. 2023
<i>Yard 3D map creation and virtual environment establishment</i> Industrial Cooperative Project funds from Daewoo Shipbuilding & Marine Engineering Co., Ltd Role - Data Acquisition, Virtual Environment Implementation, 3D Mapping	Apr. 2022 - Oct. 2022
<i>Development of AI-based transporter safety driving technology</i> Industrial Cooperative Project funds from Daewoo Shipbuilding & Marine Engineering Co., Ltd Role - Data Acquisition, Mapping, and Object Detection	Sep. 2021 - Sep. 2022

Academic Activities

<i>Reviewer, IEEE Robotics and Automation Letters (RA-L),</i>	Summer. 2025 - Present
<i>Reviewer, IEEE International Conference on Robotics and Automation (ICRA),</i>	2024

Teaching Experience

<i>Teaching Assistant</i> How to Make a Robot with Artificial Intelligence College of Engineering, Seoul National University	Fall. 2021 - Spring. 2022
<i>Teaching Assistant</i> Basic Computing: First Adventures in Computing Faculty of Liberal Education, Seoul National University	Summer. 2021 - Spring. 2022
<i>Teaching Assistant</i> Basic Mathematics and Programming Practice for Machine Learning College of Engineering, Seoul National University	Spring. 2021

Global Experience

<i>Visiting Scholar</i>	Jul. 2023 – Feb. 2024
Active Perception and Robot Intelligence Lab, University of Florida	
Advised by Dr. Shin	
Research on sonar-based underwater image super-resolution and point cloud loop detection	
<i>Exchange Student</i>	Jul. 2013 – May. 2014
University of Singapore	

Presentation

<i>The 18th KICS Future Communication Technology Workshop</i>	May 23, 2025
Seoul, Republic of Korea	
Title – Language Understanding for Spatial Understanding: Human–Robot Communication	

Awards

<i>Best Paper Award (Bronze Prize)</i>	Feb. 2023
International Conference on Electronics, Information, and Communication (ICEIC)	
Title – LoRCoN-LO: Long-Term Recurrent Convolutional Network-Based LiDAR Odometry	
<i>Outstanding Student Paper Award</i>	Jun. 2022
Summer Annual Conference of the Institute of Electronics and Information Engineers (IEIE)	
Title – Multi-Floor Mapping including the Elevator	

Work Experience

<i>Langdy</i>	Seoul, Republic of Korea
Foreign Language Education Start-Up	Jun. 2019 - Aug. 2020
Software Engineer	
Web and mobile application development	
<i>INTAKE</i>	Seoul, Republic of Korea
Food Start-Up	Jan. 2017 - Mar. 2019
Software Engineer	
Internal IT infrastructure development and high-traffic e-commerce platform management	

Skills

Programming Skills
ROS (Robot Operating Systems) 1 and 2, C++, Python, PyTorch, OpenCV, Unity
Sensors
LiDAR, Camera, IMU, GPS, Wheel Encoder
Platforms
Quadruped robot, Mobile robot, Autonomous vehicle

URLS

- Website (<https://donghwi jung.github.io>)
- Github (<https://github.com/donghwi jung>)
- Youtube (https://www.youtube.com/channel/UCqU4Gn7X3WlivtAI_yRaX0w)
- LinkedIn (<https://www.linkedin.com/in/donghwi-jung-6b9057a9/>)